CLASSIFYING FACIAL EXPRESSIONS WITH FINE-TUNED LEARNING MODELS

4AI3 PROJECT PROPOSAL
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MOTIVATIONS

- Emotional intelligence is the next step on the future development path of intelligent robots
- Robots need a reliable way to detect human emotion to:
  - understand them;
  - enable natural interaction; and
  - portray empathy
- Step 1 on the way to humanoid companions and eventually human-like partners



OBJECTIVES/ DELIVERABLES

GOAL: fine-tune a model to be able to classify **smiling, neutral,** and **frowning** faces with over *85% accuracy* in each category

- Take pictures of 75-100 people, each smiling, neutral, and frowning to build the dataset
- Process images and sort into appropriate folders
- Split data into training/validation/testing sets
- Fine-tune existing model to make accurate predictions



OBJECTIVES/ DELIVERABLES

BASIC DEMO: upload a screenshot and have the model identify the facial expression

ADVANCED DEMO: upload a short clip with multiple people and have the model identify the facial expression of each person throughout the video



Final Output Mockup:



BACKGROUND (THE MODEL)

- Requires two Ultralytic YOLO models (built off CNN)

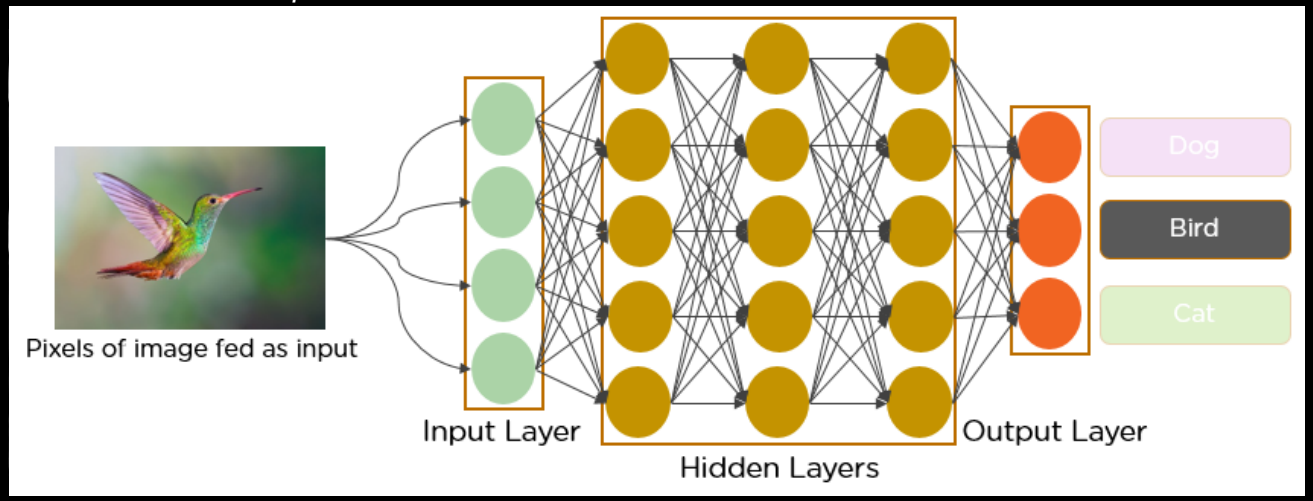
  1) *Object (face) detection*: pretrained as is

  2) **Image classification*: requires fine-tuning

- Trained on 1,281,167 224x224 pixel photos from ImageNet, currently differentiates between 1000 classes (ex. ox, canoe, tiger shark)
- Fine-tuning with our dataset replaces the final 1000-class output layer

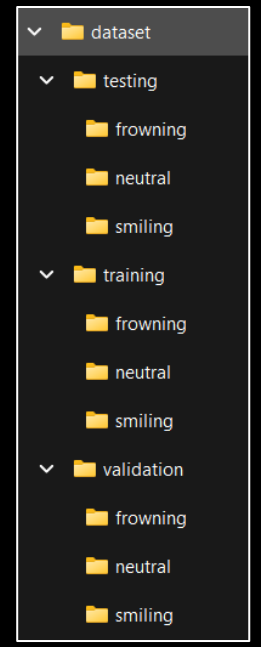


CNN Structure Example:



BACKGROUND (THE DATA)

- Data consists of 225-300 images (3 pictures from 75-100 people)
- Pre-processing involves ensuring images are valid, consistent (ie cropped correctly), and properly labeled
- YOLO model already performs color channel conversion, pixel value normalization, and resizing
- Prepare dataset by organizing into proper folders:





BACKGROUND (THE EVALUATION)

- Once fine-tuned, the model will be measured based on the following metrics:
  - Accuracy: overall share of test photos classified correctly
  - Precision *(per class)*: how many correct labels in a given image class
  - Recall *(per class)*: how many images were found and correctly labelled
  - F1 Score *(per class)*: balances the model's precision and recall
  - Confusion Matrix: compares the number of each Actual class to the number of its Predicted class



LIMITATIONS/ CHALLENGES

- Is 225-300 photos a sufficient dataset?
- Will the model have a bias towards certain physical attributes?
- How do you avoid data leakage?
- Why not use an existing dataset of faces that classifies emotions (ie. Kaggle FER-2013)?
- How accurately do facial expressions link to emotions?



TIMELINE

TOTAL PROJECT DURATION: 5 Weeks

- Week 1: data collection and setup
- Week 2: data cleanup and preprocessing
- Week 3: programming and model fine-tuning
- Week 4: final testing and demo experiments
- Week 5: create final presentation and report



REFERENCES

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